Paper B: (18 marks)
1 hour

A comprehension task. (Further details on page 72.)
Total 18 marks

Assumed Knowledge

Candidates are expected to know the content for *C1*, *C2* and *C3*.

Subject Criteria

The Units *C1* and *C2* are required for Advanced Subsidiary GCE Mathematics in order to ensure coverage of the subject criteria.

The Units *C1*, *C2*, *C3* and *C4* are required for Advanced GCE Mathematics in order to ensure coverage of the subject criteria.

Calculators

In the MEI Structured Mathematics specification, no calculator is allowed in the examination for *C1*. For all other units, including this one, a graphical calculator is allowed.

APPLICATIONS OF ADVANCED MATHEMATICS, C4		
Specification	**Ref.**	**Competence Statements**

ALGEBRA		
The general binomial expansion.	C4a1	Be able to form the binomial expansion of $(1+x)^n$ where n is any rational number and find a particular term in it.
	2	Be able to write $(a+x)^n$ in the form $a^n\left(1+\frac{x}{a}\right)^n$ prior to expansion.
Rational expressions.	3	Be able to simplify rational expressions.
Partial fractions.	4	Be able to solve equations involving algebraic fractions.
	5	Know how to express algebraic fractions as partial fractions.
	6	Know how to use partial fractions with the binomial expansion to find the power series for an algebraic fraction.

TRIGONOMETRY		
sec, cosec and cot.	C4t1	Know the definitions of the sec, cosec and cot functions.
	2	Understand the relationship between the graphs of the sin, cos, tan, cosec, sec and cot functions.
	3	Know the relationships $\tan^2\theta+1=\sec^2\theta$ and $\cot^2\theta+1=\mathrm{cosec}^2\theta$.
Compound angle formulae.	4	Be able to use the identities for $\sin(\theta\pm\phi)$, $\cos(\theta\pm\phi)$, $\tan(\theta\pm\phi)$.
	5	Be able to use identities for $\sin 2\theta$, $\cos 2\theta$ (3 versions), $\tan 2\theta$.
Solution of trigonometrical equations.	6	Be able to solve simple trigonometrical equations within a given range including the use of any of the trigonometrical identities above.
	7	Know how to write the function $a\cos\theta \pm b\sin\theta$ in the forms $R\sin(\theta\pm\alpha)$ and $R\cos(\theta\pm\alpha)$ and how to use these to sketch the graph of the function, find its maximum and minimum values and to solve equations.

PARAMETRIC EQUATIONS		
The use of parametric equations.	C4g1	Understand the meaning of the terms parameter and parametric equations.
	2	Be able to find the equivalent cartesian equation for parametric equations.
	3	Recognise the parametric form of a circle.
	4	Be able to find the gradient at a point on a curve defined in terms of a parameter by differentiation.

APPLICATIONS OF ADVANCED MATHEMATICS, C4			
Ref.	**Notes**	**Notation**	**Exclusions**

ALGEBRA

C4a1	For $\lvert x \rvert < 1$ when n is not a positive integer.		
2	$\left\lvert \frac{x}{a} \right\rvert < 1$ when n is not a positive integer.		
3	Including factorising, cancelling and algebraic division.		
4			
5	Proper fractions with the following denominators $(ax+b)(cx+d)$ $(ax+b)(cx+d)^2$ $(ax+b)(x^2+c^2)$		Improper fractions.
6			

TRIGONOMETRY

C4t1	Including knowledge of the angles for which they are undefined.		
2			
3			
4			
5			
6	Including identities from earlier units. Knowledge of principal values.		
7			

PARAMETRIC EQUATIONS

C4g1			
2			
3			
4	$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)}{\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)}$ Use to find the equations of tangents and normals to a curve.		Stationary points.

APPLICATIONS OF ADVANCED MATHEMATICS, C4		
Specification	**Ref.**	**Competence Statements**

		CALCULUS
Numerical integration.	C4c1	Be able to use the trapezium rule to find an integral to a given level of accuracy.
Partial fractions.	2	Be able to use the method of partial fractions in integration.
Volumes of revolution.	3	Be able to calculate the volumes of the solids generated by rotating a plane region about the x-axis or the y-axis.
Differential equations.	4	Be able to formulate first order differential equations.
	5	Be able to solve first order differential equations.
		VECTORS
Vectors in two and three dimensions.	C4v1	Understand the language of vectors in two and three dimensions.
	2	Be able to add vectors, multiply a vector by a scalar, and express a vector as a combination of others.
The scalar product.	3	Know how to calculate the scalar product of two vectors, and be able to use it to find the angle between two vectors.
Coordinate geometry in two and three dimensions.	4	Be able to find the distance between two points, the midpoint and other points of simple division of a line.
The equations of lines and planes.	5	Be able to form and use the equation of a line.
	6	Be able to form and use the equation of a plane.
The intersection of a line and a plane.	7	Know that a vector which is perpendicular to a plane is perpendicular to any line in the plane.
	8	Know that the angle between two planes is the same as the angle between their normals.
	9	Be able to find the intersection of a line and a plane.
		COMPREHENSION
The ability to read and comprehend a mathematical argument or an example of the application of mathematics.	C4p1	Be able to follow mathematical arguments and descriptions of the solutions of problems when given in writing.
	2	Understand the modelling cycle and realise that it can be applied across many branches of mathematics.

APPLICATIONS OF ADVANCED MATHEMATICS, C4			
Ref.	**Notes**	**Notation**	**Exclusions**

CALCULUS			
C4c1	Use of increasing numbers of strips to improve the accuracy. Use of increasing numbers of strips to estimate the error.		Questions requiring more than 3 applications of the trapezium rule.
2			
3			Axes of rotation other than the *x*- and *y*-axes.
4	From given information about rates of change.		
5	Differential equations with separable variables only.		
VECTORS			
C4v1	Scalar, vector, modulus, magnitude, direction, position vector, unit vector, cartesian components, equal vectors, parallel vectors.	$\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$, $\hat{\mathbf{r}}$ $\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$	
2	Geometrical interpretation.		
3	Including test for perpendicular vectors. The angle between two lines.	$\mathbf{a}.\mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3$ $= \lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta$	
4			
5	In vector and cartesian form.	Line: $\mathbf{r} = \mathbf{a} + t\mathbf{u}$ $\frac{x-a_1}{u_1} = \frac{y-a_2}{u_2} = \frac{z-a_3}{u_3} (= t)$.	
6	In vector and cartesian form.	Plane: $(\mathbf{r} - \mathbf{a}).\mathbf{n} = 0$ $n_1x + n_2y + n_3z + d = 0$ where $d = -\mathbf{a}.\mathbf{n}$.	
7	If a vector is perpendicular to two non-parallel lines in a plane, it is perpendicular to the plane.		
8			
9			
COMPREHENSION			
C4p1	This may be assessed using a real world modelling context.		
2	Abstraction from a real-world situation to a mathematical description; approximation simplification and solution; check against reality; progressive refinement.		

Applications of Advanced Mathematics (C4) Comprehension Task

Rationale

The aim of the comprehension task is to foster an appreciation among students that, in learning mathematics, they are acquiring skills which transcend the particular items of the specification content which have made up their course.

The objectives are that students should be able to:

- read and comprehend a mathematical argument or an example of the application of mathematics;
- respond to a synoptic piece of work covering ideas permeating their whole course;
- appreciate the relevance of particular techniques to real-world problems.

Description and Conduct

Paper B of *Applications of Advanced Mathematics (C4)* consists of a comprehension task on which candidates are expected to take no more than 40 minutes. The task takes the form of a written article followed by questions designed to test how well candidates have understood it. Care will be taken in preparing the task to ensure that the language is readily accessible.

Candidates are allowed to bring standard English dictionaries into the examination. Full regulations can be found in the JCQ booklet *Instructions for conducting examinations*, published annually.

The use of bi-lingual translation dictionaries by candidates for whom English is not their first language has to be applied for under the access arrangements rules. Full details can be found in the JCQ booklet *Access Arrangements, Reasonable Adjustments and Special Consideration*, published annually.

Content

By its nature, the content of the written piece of mathematics cannot be specified in the detail of the rest of the specification. However knowledge of GCSE and *C1*, *C2* and *C3* will be assumed, as well as the content of the rest of this unit. Candidates are expected to be aware of ideas concerning accuracy and errors. The written piece may follow a modelling cycle and in that case candidates will be expected to recognise it. No knowledge of mechanics will be assumed.

6.5 FURTHER CONCEPTS FOR ADVANCED MATHEMATICS, FP1 (4755) AS

Objectives

To develop an understanding of the rigour and technical accuracy needed for more advanced study of mathematics.

Assessment

Examination: (72 marks)
1 hour 30 minutes.
The examination paper has two sections.

Section A: 5-7 questions, each worth at most 8 marks.
Section Total: 36 marks.

Section B: three questions, each worth about 12 marks.
Section Total: 36 marks.

Assumed Knowledge

Candidates are expected to know the content for *C1* and *C2*.

Subject Criteria

This unit is required for Advanced Subsidiary Further Mathematics. Candidates proceeding to Advanced GCE Further Mathematics will also need *FP2*.

The Units *C1*, *C2*, *C3* and *C4* are required for Advanced GCE Mathematics in order to ensure coverage of the subject criteria.

Calculators

In the MEI Structured Mathematics specification, no calculator is allowed in the examination for *C1*. For all other units, including this one, a graphical calculator is allowed.

FURTHER CONCEPTS FOR ADVANCED MATHEMATICS, FP1		
Specification	**Ref.**	**Competence Statements**

COMPLEX NUMBERS		
Quadratic equations.	FP1j1	Be able to solve any quadratic equation with real coefficients.
Addition, subtraction, multiplication and division of complex numbers.	2	Understand the language of complex numbers.
	3	Be able to add, subtract, multiply and divide complex numbers given in the form: $x + y\mathrm{j}$, where *x* and *y* are real.
	4	Know that a complex number is zero if and only if both the real and imaginary parts are zero.
Application of complex numbers to the solution of polynomial equations with real coefficients.	5	Know that the complex roots of real polynomial equations with real coefficients occur in conjugate pairs.
	6	Be able to solve equations of higher degree with real coefficients in simple cases.
	7	Know how to represent complex numbers and their conjugates on an Argand diagram.
	8	Be able to represent the sum and difference of two complex numbers on an Argand diagram.
Modulus-argument form	9	Be able to represent a complex number in modulus-argument form.
Simple loci in the Argand diagram.	10	Be able to represent simple sets of complex numbers as loci in the Argand diagram.

CURVE SKETCHING		
Treatment and sketching of graphs of rational functions.	FP1C1	Be able to sketch the graph of $y = \mathrm{f}(x)$ obtaining information about symmetry, asymptotes parallel to the axes, intercepts with the coordinate axes, behaviour near $x = 0$ and for numerically large *x*.
	2	Be able to ascertain the direction from which a curve approaches an asymptote.
	3	Be able to use a curve to solve an inequality.

FURTHER CONCEPTS FOR ADVANCED MATHEMATICS, FP1			
Ref.	**Notes**	**Notation**	**Exclusions**

COMPLEX NUMBERS			
FP1j1		$j^2 = -1$	
2	Real part, imaginary part, complex conjugate, modulus, argument, real axis, imaginary axis.	$z = x + y\text{j}$ $z^* = x - y\text{j}$ $\text{Re}(z) = x$ $\text{Im}(z) = y$	
3	Division using complex conjugates.		
4			
5	e.g. to solve a cubic equation given one complex root.		Equations with degree > 4. Equations with more than 2 complex roots (unless they are purely imaginary).
6			
7			
8			
9	Conversion between the forms $z = x + y\text{j}$ and $z = r(\cos\theta + \sin\theta\,\text{j})$.	Modulus of z: $\lvert z \rvert$ Radian measure.	
10	Circles of the form $\lvert z - a \rvert = r$. Half lines of the form $\arg(z - a) = \theta$.		

CURVE SKETCHING			
FP1C1	Cases where $\text{f}(x) = \frac{\text{g}(x)}{\text{h}(x)}$ and the order of $\text{g}(x) \leq$ the order of $\text{h}(x)$.		Oblique asymptotes.
2	Vertical and horizontal asymptotes.		Formal treatment.
3			

FURTHER CONCEPTS FOR ADVANCED MATHEMATICS, FP1		
Specification	**Ref.**	**Competence Statements**

PROOF		
Meaning of the terms if, only if, necessary and sufficient.	FP1p1	Be able to use the terms *if*, *only if*, *necessary* and *sufficient* correctly in any appropriate context.
Identities.	2	Know the difference between an equation and an identity.
	3	Be able to find unknown constants in an identity.
Proof by induction.	4	Be able to construct and present a correct proof using mathematical induction.

ALGEBRA		
Summation of simple finite series.	FP1a1	Know the difference between a sequence and a series.
	2	Be able to sum a simple series.
	3	Know the meaning of the word *converge* when applied to either a sequence or a series.
The manipulation of simple algebraic inequalities.	4	Be able to manipulate simple algebraic inequalities, to deduce the solution of such an inequality.
Relations between the roots and coefficients of quadratic, cubic and quartic equations.	5	Appreciate the relationship between the roots and coefficients of quadratic, cubic and quartic equations.
	6	Be able to form a new equation whose roots are related to the roots of a given equation by a linear transformation.

FURTHER CONCEPTS FOR ADVANCED MATHEMATICS, FP1			
Ref.	**Notes**	**Notation**	**Exclusions**
PROOF			
FP1p1			
2	An identity is true for all values of the variables, an equation for only certain values.	$\equiv$	
3	By substituting particular values for the variable.		
4	Proofs of the sums of simple series. The result to be proved will always be given explicitly.		Proofs of divisibility.
ALGEBRA			
FP1a1			
2	Using standard formulae for Σr, Σr^2 and Σr^3. Using the method of differences.		Derivation of partial fractions.
3			
4	Including those expressible in the form $\mathrm{f}(x) > 0$ where $\mathrm{f}(x)$ can be expressed as a product of linear factors.		Inequalities involving algebraic fractions where the numerator and denominator are of order ≥ 2.
5		Roots $\alpha, \beta, \gamma, \delta$.	Equations of degree ≥ 5.
6			New equations with non-linear combinations of roots.

FURTHER CONCEPTS FOR ADVANCED MATHEMATICS, FP1		
Specification	**Ref.**	**Competence Statements**

MATRICES		
Matrix addition and multiplication.	FP1m1	Be able to add, subtract and multiply conformable matrices, and to multiply a matrix by a scalar.
	2	Know the zero and identity matrices, and what is meant by equal matrices.
	3	Know that matrix multiplication is associative but not commutative.
Linear transformations in a plane and their associated 2x2 matrices.	4	Be able to find the matrix associated with a linear transformation and vice-versa.
Combined transformations in a plane.	5	Understand successive transformations and the connection with matrix multiplication.
Invariance.	6	Understand the meaning of invariant points and lines of invariant points in a plane and how to find them.
Determinant of a matrix.	7	Be able to find the determinant of a 2x2 matrix.
	8	Know that the determinant gives the area scale factor of the transformation, and understand the significance of a zero determinant.
The meaning of the inverse of a square matrix.	9	Understand what is meant by an inverse matrix.
	10	Be able to find the inverse of a non-singular 2x2 matrix.
The product rule for inverses.	11	Appreciate the product rule for inverse matrices.
Solution of equations.	12	Know how to use matrices to solve linear equations.
	13	In the case of 2 linear equations in 2 unknowns, be able to give a geometrical interpretation of a case where the matrix is singular.

FURTHER CONCEPTS FOR ADVANCED MATHEMATICS, FP1			
Ref.	**Notes**	**Notation**	**Exclusions**

MATRICES

FP1m1	Matrices of any suitable order.	$\mathbf{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$	
2		$\mathbf{O}$ (zero) $\mathbf{I}$ (identity).	
3			
4	Reflection in the x and y axes and in the lines $y = \pm x$. Rotation centre the origin through an angle θ. Enlargement centre the origin.	Column vectors for the position of points.	
5			
6			
7		$\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ or det $\mathbf{M}$.	
8	The terms *singular* and *non-singular*.		
9	Square matrices of any order.	$\mathbf{M}^{-1}$ (inverse).	
10	Candidates should know that: $\mathbf{AB} = k\mathbf{I} \Rightarrow \mathbf{A}^{-1} = \frac{1}{k}\mathbf{B}$.	The inverse of $\mathbf{M}$ is $\frac{1}{\det \mathbf{M}} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$	
11	$(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$.		
12			Equations with 3 or more unknowns when the inverse matrix is not known.
13	The graphs of the equations are parallel lines.		